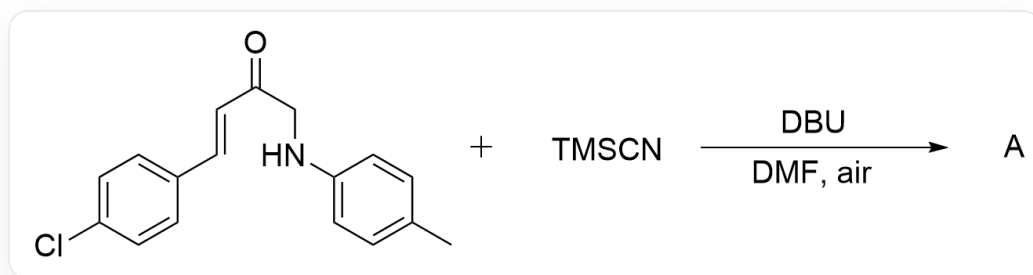


题目

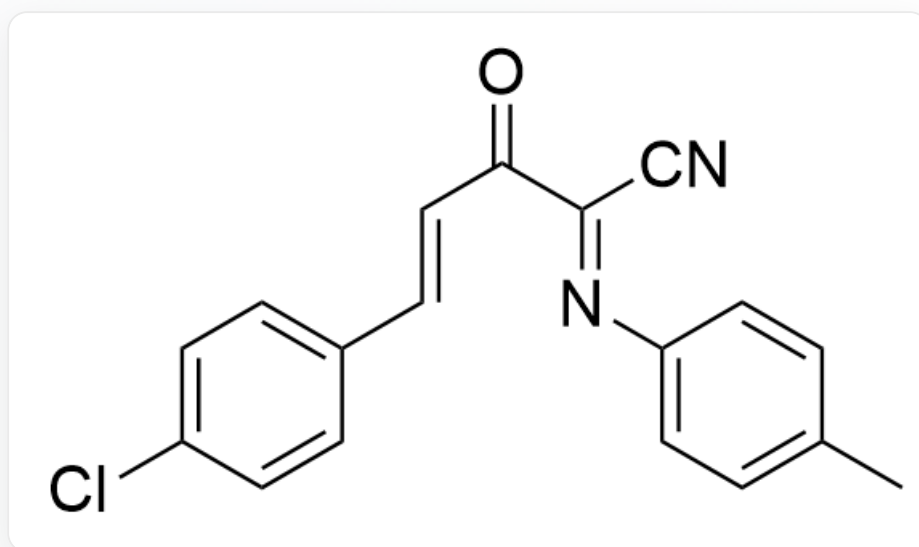


O=C(CNC1=CC=C(C)C=C1)/C=C/C2=CC=C(Cl)C=C2.C[Si](C)(C#N)C>[DBU],[DMF]>[A], A为反应产物, 应在空气中进行

已知该反应包含自由基转化过程。请尝试预测该反应的主产物 **A** 以及不含氰基的副产物 **B** (分子式: $C_{17}H_{14}ClNO_2$) 的结构式

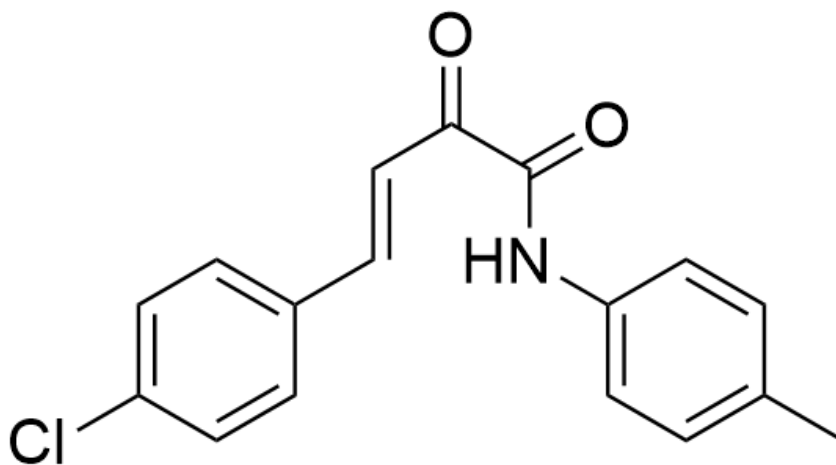
A. 其他选项均不正确

B.



O=C(/C(C#N)=N/C1=CC=C(C)C=C1)/C=C/C2=CC=C(Cl)C=C2, 主产物 **A**

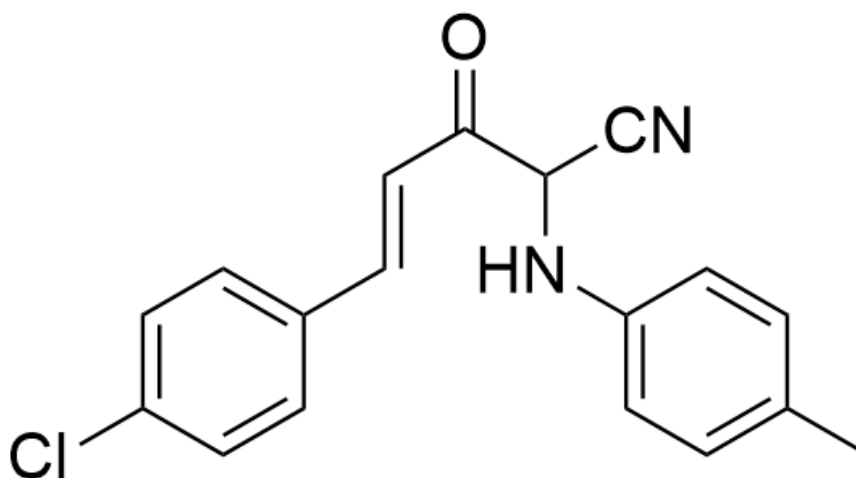
主产物 **A**



O=C(C(NC1=CC=C(C)C=C1)=O)/C=C/C2=CC=C(C)C=C2,副产物**B**

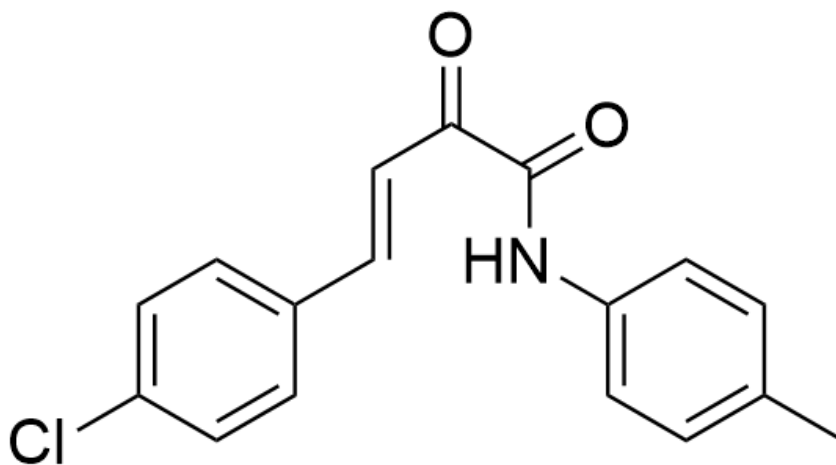
副产物 **B**

C.



O=C(C(C#N)NC1=CC=C(C)C=C1)/C=C/C2=CC=C(C)C=C2,主产物**A**

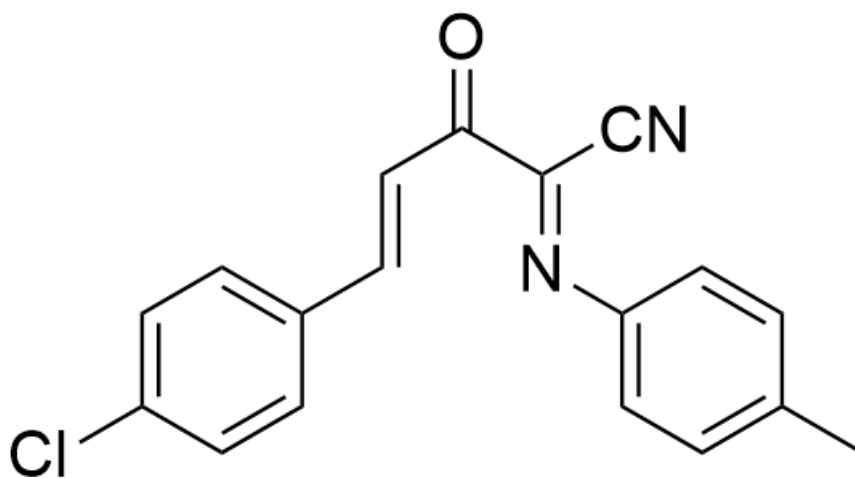
主产物 **A**



O=C(C(NC1=CC=C(C)C=C1)=O)/C=C/C2=CC=C(Cl)C=C2,副产物B

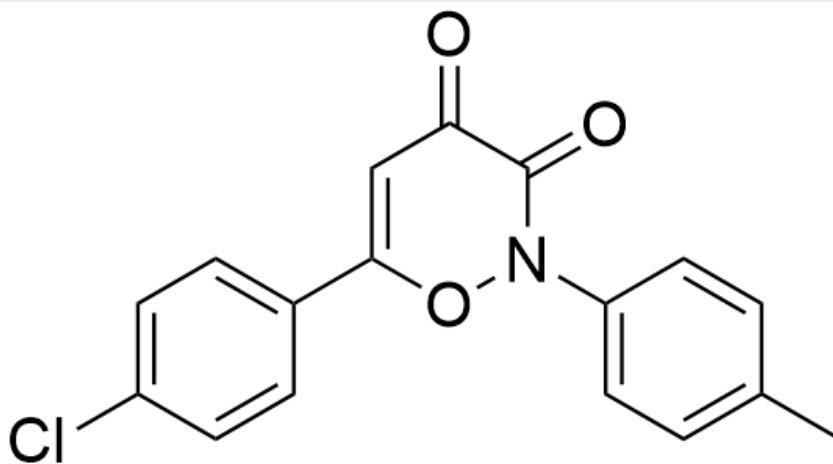
副产物 B

D.



O=C(/C(C#N)=N/C1=CC=C(C)C=C1)/C=C/C2=CC=C(Cl)C=C2,主产物A

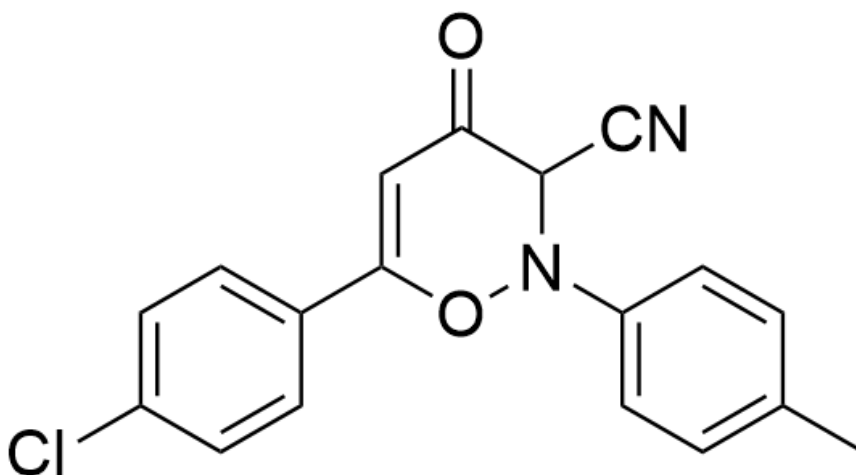
主产物 A



O=C(C(N(O1)C2=CC=C(C)C=C2)=O)C=C1C3=CC=C(Cl)C=C3,副产物B

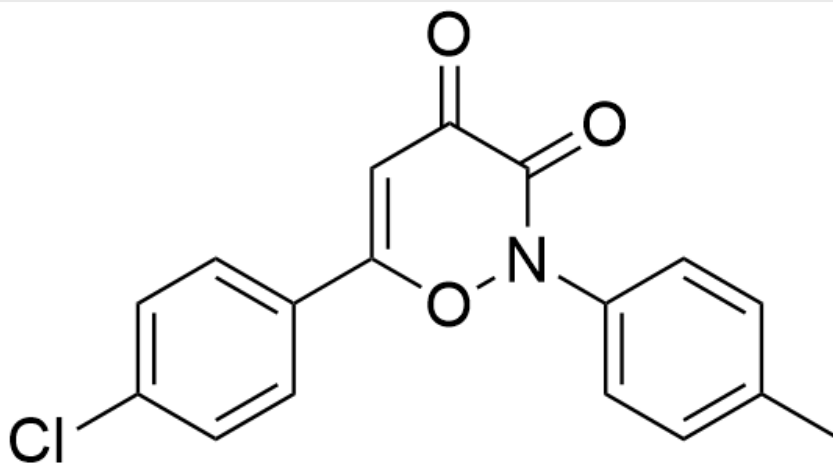
副产物 B

E.



O=C(C(C#N)N(O1)C2=CC=C(C)C=C2)C=C1C3=CC=C(Cl)C=C3,主产物A

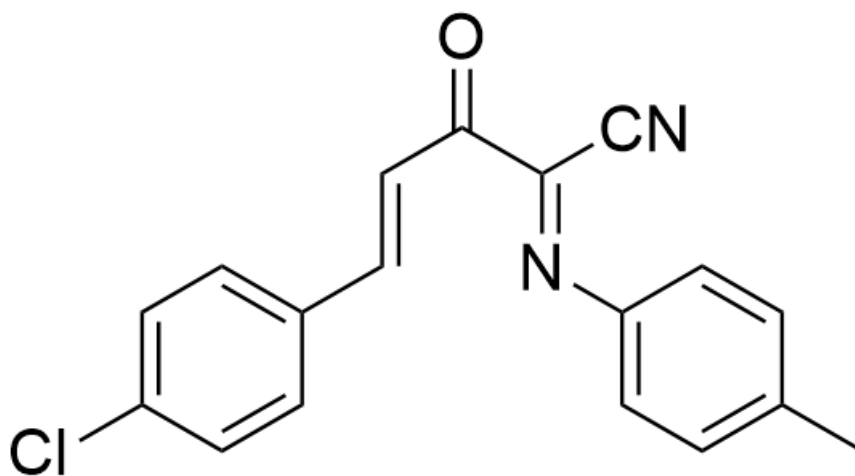
主产物 A



O=C(C(N(O1)C2=CC=C(C)C=C2)=O)C=C1C3=CC=C(Cl)C=C3,副产物**B**

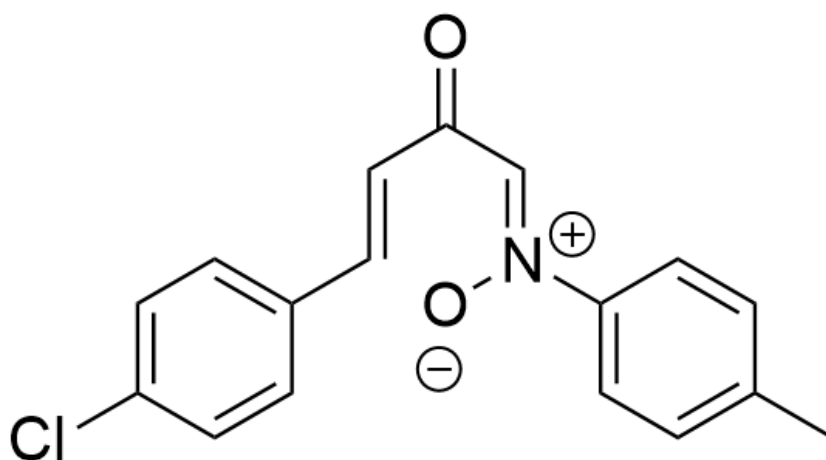
副产物 **B**

F.



O=C(/C(C#N)=N/C1=CC=C(C)C=C1)/C=C/C2=CC=C(Cl)C=C2,主产物**A**

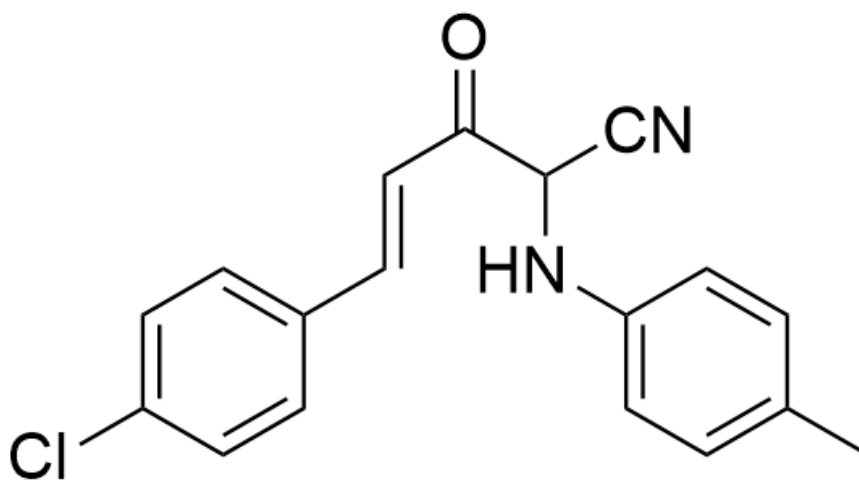
主产物 **A**



O=C(/C=[N+](O-)/C1=CC=C(C)C=C1)/C=C/C2=CC=C(Cl)C=C2,副产物B

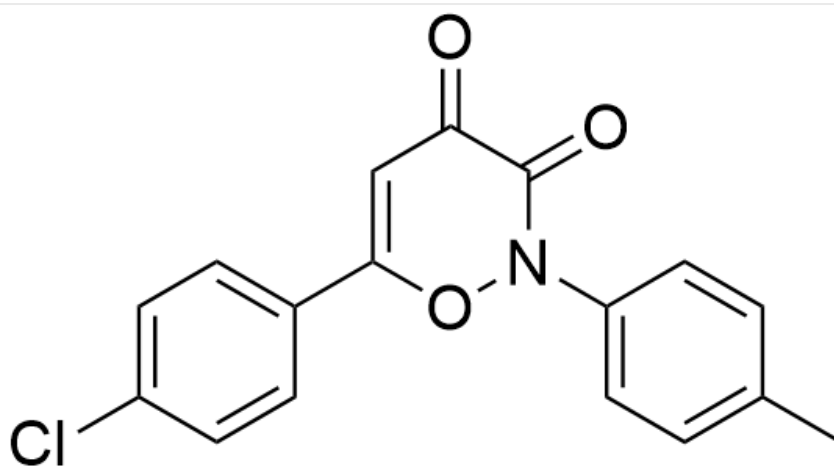
副产物 B

G.



O=C(C(C#N)NC1=CC=C(C)C=C1)/C=C/C2=CC=C(Cl)C=C2,主产物A

主产物 A



O=C(C(N(O1)C2=CC=C(C(C)=C2)=O)C=C1C3=CC=C(C(Cl)=C3)C=C3,副产物**B**

副产物**B**

答案

正确答案: **B**

详细解析

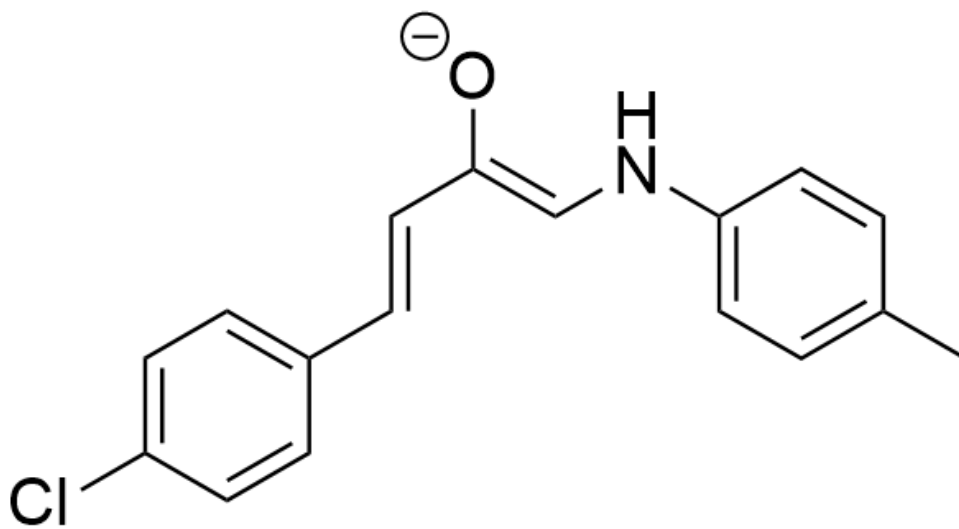
根据副产物的化学式可知生成副产物的过程中发生了氧化，碳数不变而多了一个氧，结合反应条件可知氧供体只能是来自空气的氧气，而氧气便可作为自由基引发剂引发题干所说的自由基转化过程。

CHECKPOINT

1 PTS

氧气作为自由基引发剂

首先底物在碱DBU的作用下去质子形成中间体1



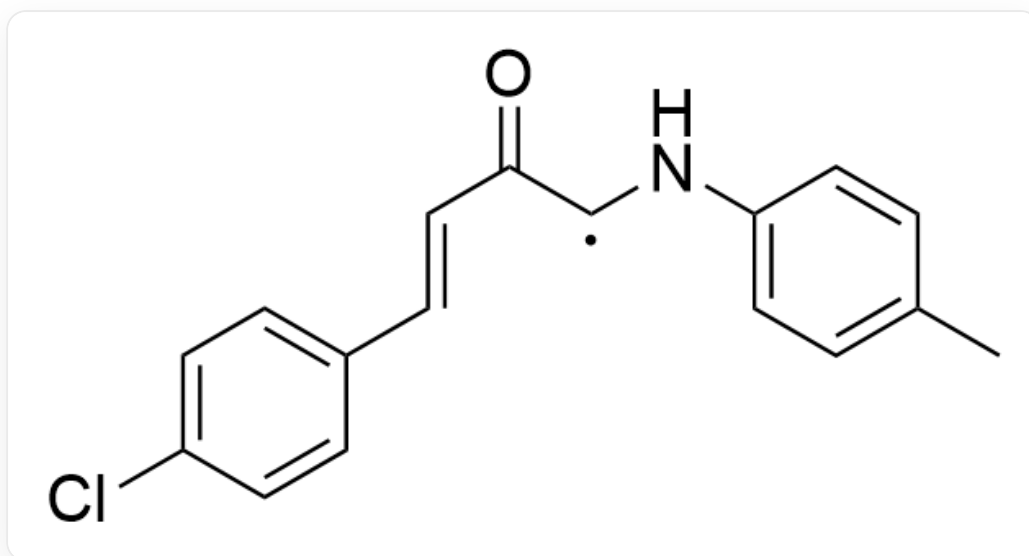
[O-]C(/C=C/C1=CC=C(Cl)C=C1)=C\NC2=CC=C(C)C=C2

CHECKPOINT

1 PTS

[O-]C(/C=C/C1=CC=C(Cl)C=C1)=C\NC2=CC=C(C)C=C2

随后被氧气夺取一个电子形成超氧阴离子自由基和自由基中间体2



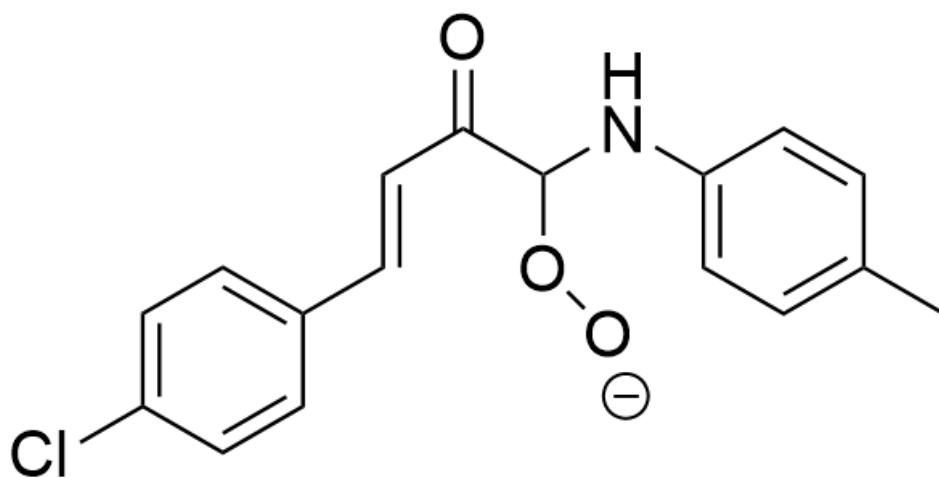
O=C(C([X])NC1=CC=C(C)C=C1)/C=C/C2=CC=C(Cl)C=C2, X是单电子

CHECKPOINT

1 PTS

O=C(C([X])NC1=CC=C(C)C=C1)/C=C/C2=CC=C(Cl)C=C2, X是单电子

该自由基与超氧阴离子自由基结合得到中间体3



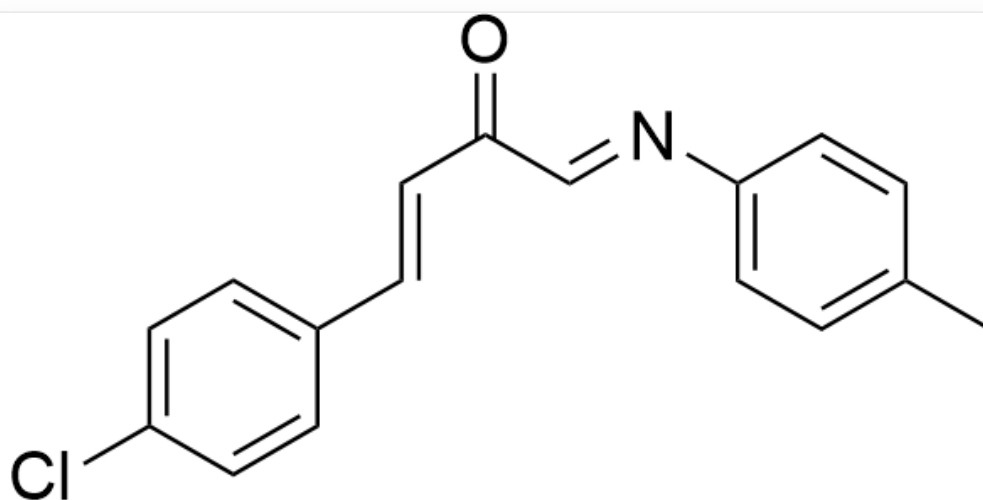
O=C(C(O[O-])NC1=CC=C(C)C=C1)/C=C/C2=CC=C(Cl)C=C2

CHECKPOINT

1 PTS

O=C(C(O[O-])NC1=CC=C(C)C=C1)/C=C/C2=CC=C(Cl)C=C2

随后发生消除反应得到中间体4



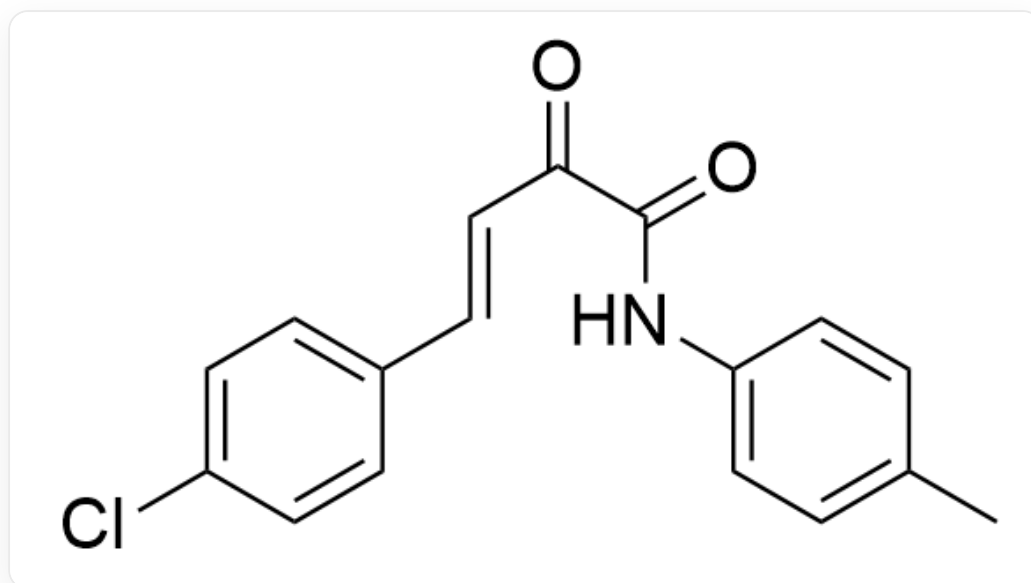
O=C(/C=N/C1=CC=C(C)C=C1)/C=C/C2=CC=C(Cl)C=C2

CHECKPOINT

1 PTS

O=C(/C=N/C1=CC=C(C)C=C1)/C=C/C2=CC=C(Cl)C=C2

此处如果只消除1分子 H₂O 的话，将得到副产物 **B**，这也符合题目给出的化学式。



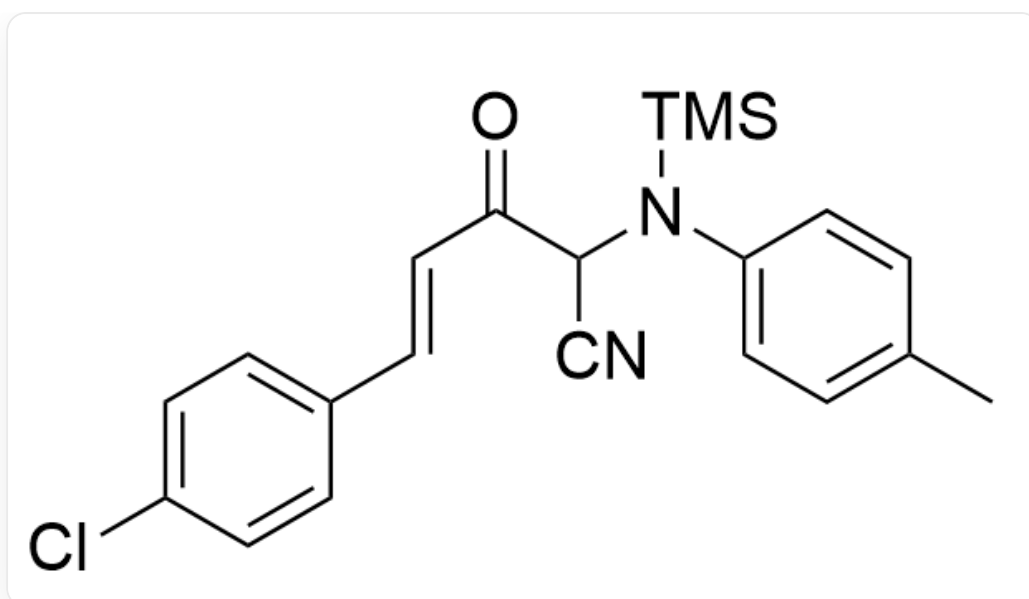
O=C(C(NC1=CC=C(C)C=C1)=O)/C=C/C2=CC=C(Cl)C=C2,副产物 **B**

CHECKPOINT

1 PTS

O=C(C(NC1=CC=C(C)C=C1)=O)/C=C/C2=CC=C(Cl)C=C2,副产物 $\mathbf{B}$

中间体4与 TMS-CN 发生快速的加成反应得到中间体5



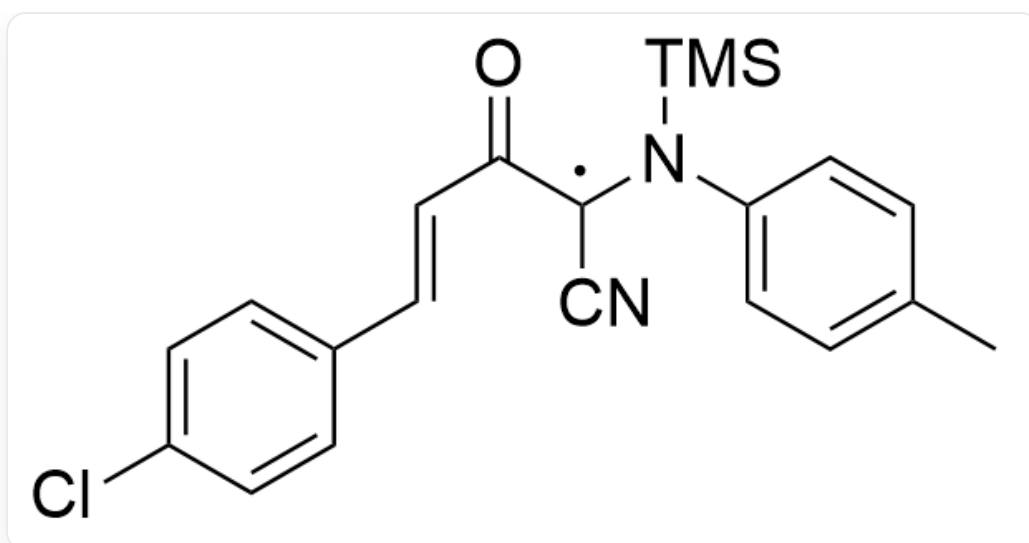
O=C(C(C#N)N([Si](C)(C)C)C1=CC=C(C)C=C1)/C=C/C2=CC=C(Cl)C=C2

CHECKPOINT

1 PTS

O=C(C(C#N)N([Si](C)(C)C)C1=CC=C(C)C=C1)/C=C/C2=CC=C(Cl)C=C2

该中间体5进一步被氧气夺去1个电子形成自由基中间体6



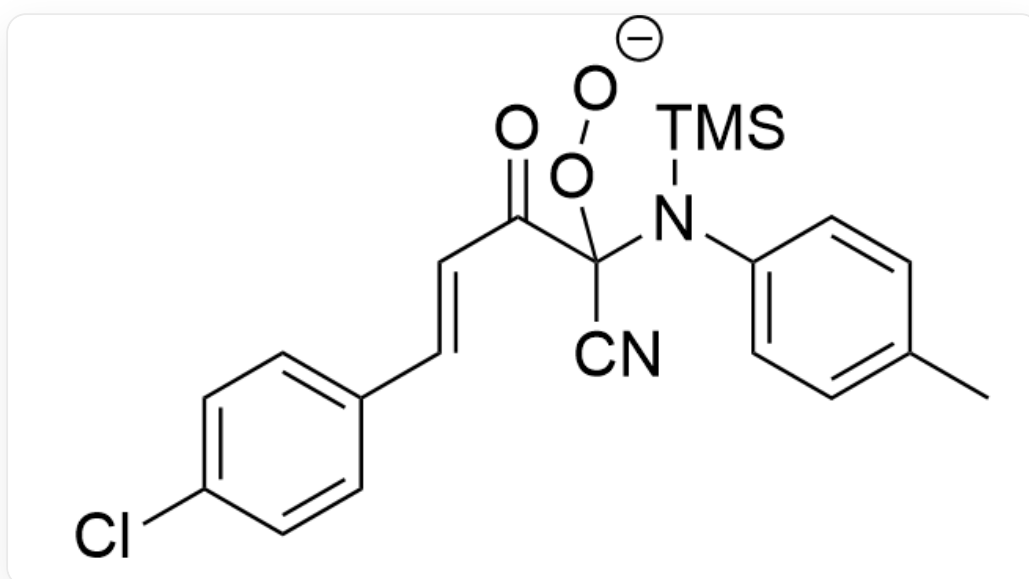
O=C(C([X])(C#N)N([Si](C)(C)C)C1=CC=C(C)C=C1)/C=C/C2=CC=C(Cl)C=C2, X为单电子

CHECKPOINT

1 PTS

O=C(C([X])(C#N)N([Si](C)(C)C)C1=CC=C(C)C=C1)/C=C/C2=CC=C(Cl)C=C2, X为单电子

该自由基中间体与超氧阴离子自由基结合得到中间体7



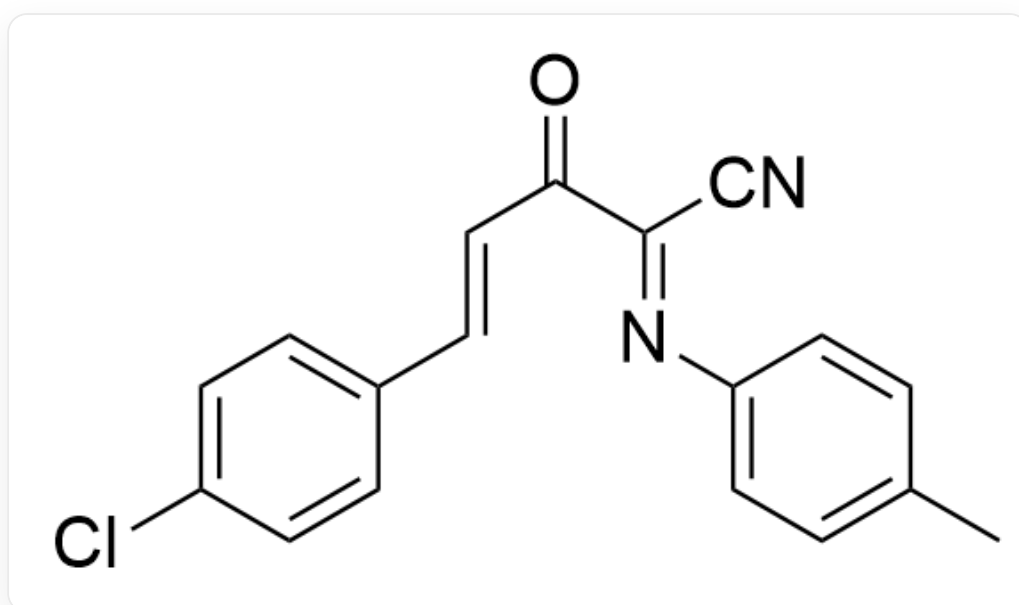
O=C(C(N([Si](C)(C)C)C1=CC=C(C)C=C1)([O-][O-])C#N)/C=C/C2=CC=C(Cl)C=C2

CHECKPOINT

1 PTS

O=C(C(N([Si](C)(C)C)C1=CC=C(C)C=C1)(O[O-])C#N)/C=C/C2=CC=C(Cl)C=C2

最后脱去一分子 TMS-超氧阴离子得到主产物 A



O=C(/C(C#N)=N/C1=CC=C(C)C=C1)/C=C/C2=CC=C(Cl)C=C2,主产物 A

CHECKPOINT

1 PTS

O=C(/C(C#N)=N/C1=CC=C(C)C=C1)/C=C/C2=CC=C(Cl)C=C2,主产物 A